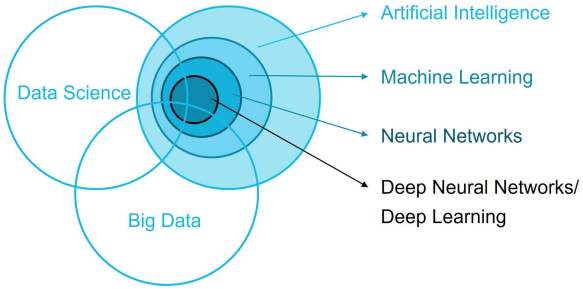
Understanding AI and Data Science

Dr. Shailesh Sivan

CONFUSED ?



COMPUTER SYSTEMS

- An electronics device
- Capable of storing and processing data
- Can be used for controlling other devices

One which do computing

ROBOTS AND COMPUTER

Are Computers Intelligent or Dumb ?

COMPUTERS ARE DUMB ⊗ !

- Computer is not a magical device.
- It performs only those works which man can do!
- But with very high speed and reliable accuracy.
- It has no intelligence quality or thinking power

DATA, INFORMATION & KNOWLEDGE

Elephant

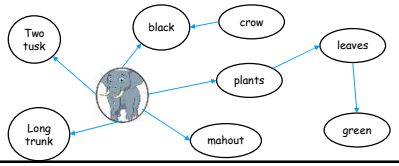
herbivrous
big trunk tusk
mahout
black

Data

- Biggest animal in the land
- Has two tusk
- Black in Colour
- Has a long trunk
- Eats plants
- Mahout is the keeper

Information

Knowledge



ARTIFICIAL INTELLIGENCE

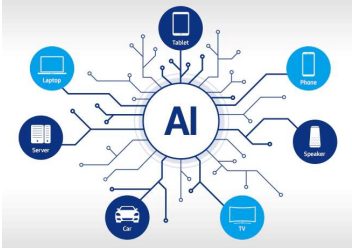
Can We make Computers Intelligent ?

Yeah Of course !

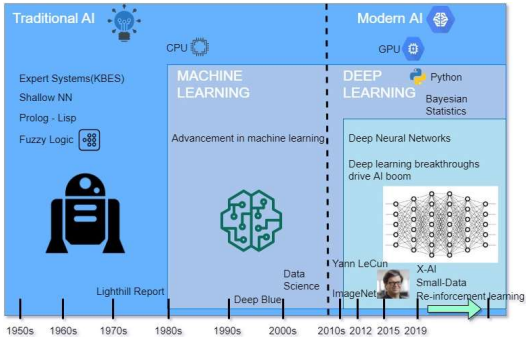
+ Knowledge = **Artificial Intelligence**

Artificial Intelligence is a scientific domain that deals with **making intelligence** for systems with the help of **data** and **algorithms** to enable them for **decision making**, **optimization**, and **content generation**

To make a system "**think like a human**"



ARTIFICIAL INTELLIGENCE



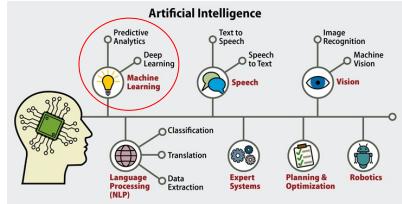
ARTIFICIAL INTELLIGENCE

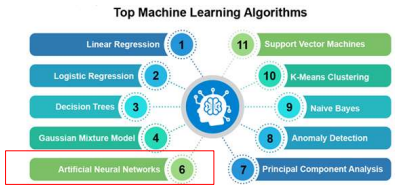
Types of AI

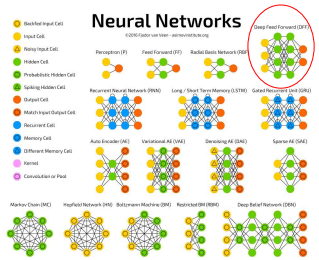
Narrow Artificial Intelligence
able to perform only one specific task

General Artificial Intelligence
able to perform various tasks like a human

Artificial Super Intelligence
AI that surpasses human abilities

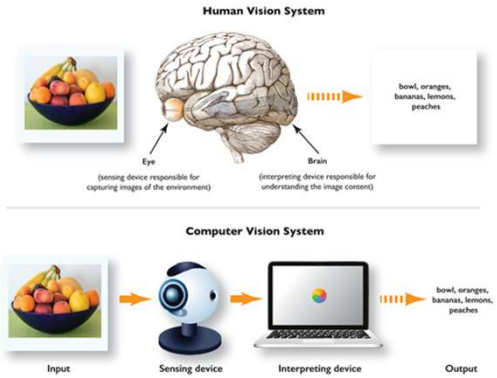






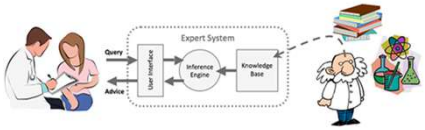
COMPUTER VISION

"How Machine See ?"



Expert Systems

An Expert System is a computer program designed to act as an expert in a particular domain (area of expertise).



Expert systems currently are designed to assist experts, not to replace them.

They have been used in medical diagnosis, chemical analysis, geological explorations etc.

ROBOTS : AS INTELLIGENT MACHINES

Robots are mechatronic(**mechanical** + **electronic**) machines that can sense, plan, and act according to programmed logic(**artificial intelligence**)

WHAT ROBOTS CAN DO ?

Make Pancakes

Works in a Factory

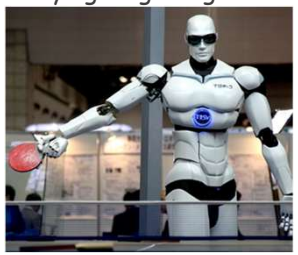
Draw portraits

Robotic arms

HUMANOIDS

Home Helper

Playing Ping-Pong



Personal Assistant

ROBOT VEHICLES

Space Exploration

Vacums

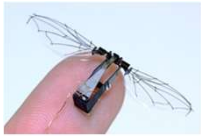
Carriers

BIOMETRIC ROBOTS

Soar like a bird

Water Pollution Monitoring

Swarms

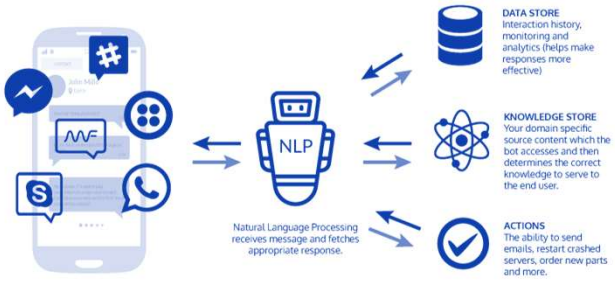


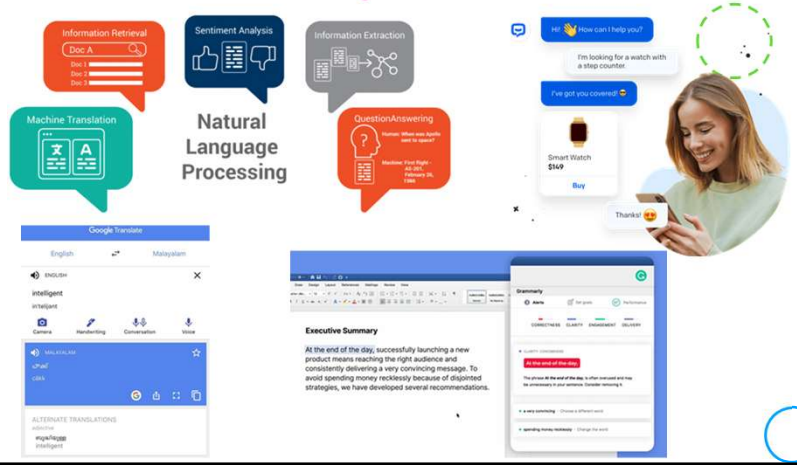
Robot Dog

NATURAL LANGUAGE PROCESSING

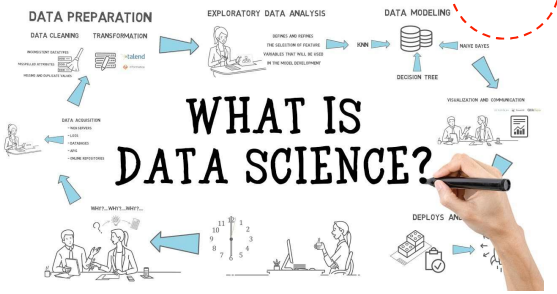
"How Machine Communicate in Human Language ?"

NATURAL LANGUAGE PROCESSING(NLP)

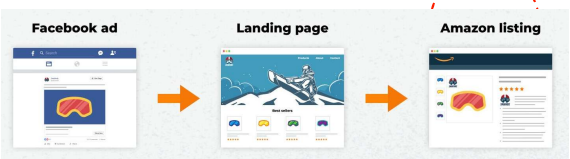


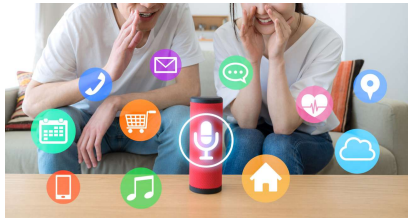


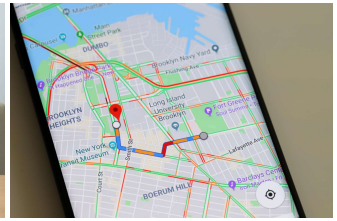
DATA SCIENCE



IS DATA IMPORTANT?







WHY DATA IS IMPORTANT ?

- Make Informed Decisions
- Get The Results You Want
- Find Solutions To Problems
- Back Up Your Arguments
- Be Strategic In Your Approaches
- Keep Track Of It All
- Know What You Are Doing Well

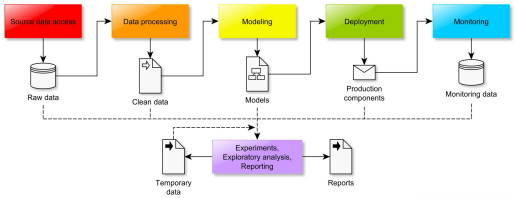
DATA IS NEW OIL

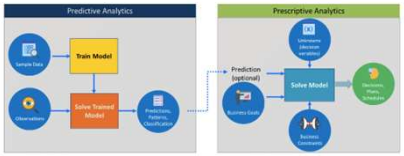
REALITY ABOUT DATA

SOLUTION ?

We are now **drowing** in data ! but **starving** for insights ☹

Data Science Pipeline





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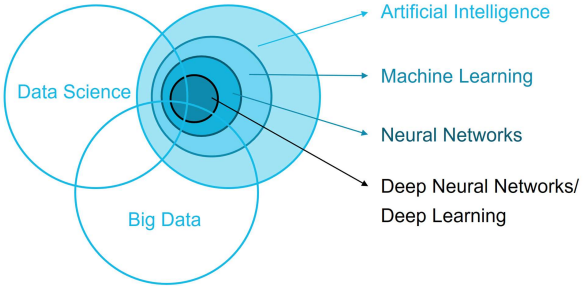
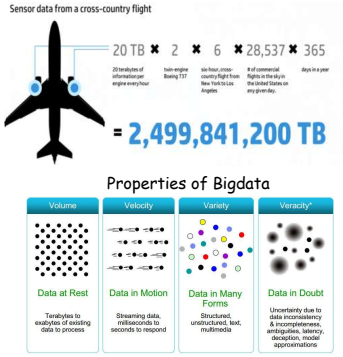
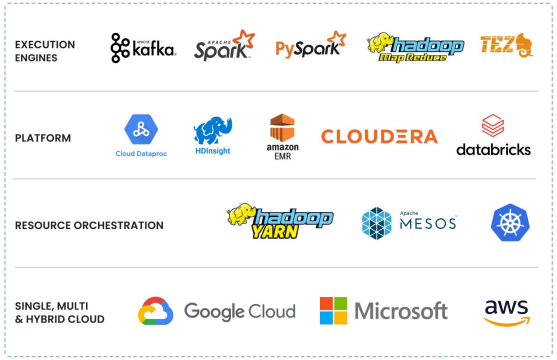


Big Data



Is Big Data = Large Data ?

Big Data



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